

HIGH PERFORMANCE COMPUTING

Tugas 3



Disusun oleh:

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FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM

Program Studi S-1 Teknik Informatika

2026

Source Code: <https://github.com/WilsonnnTan/hpc-assignment/tree/master/assignment-3>

1. Start the cluster

```
wilson@DESKTOP-KSAB409:/mnt/c/College_Computer_Science/Sem 7/high-performance-computing$ cd assignment-3
wilson@DESKTOP-KSAB409:/mnt/c/College_Computer_Science/Sem 7/high-performance-computing/assignment-3$ docker compose up -d --build
[+] Building 3.9s (13/13) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 622B                                          0.0s
=> [internal] load build definition from Dockerfile                    0.0s
=> => transferring dockerfile: 653B                                    0.0s
=> [internal] load metadata for docker.io/library/ubuntu:22.04        2.7s
=> [auth] library/ubuntu:pull token for registry-1.docker.io          0.0s
=> [internal] load .dockerignore                                       0.0s
=> => transferring context: 2B                                         0.0s
=> [1/5] FROM docker.io/library/ubuntu:22.04@sha256:5ec03bb3441e8b0bf3b4f9cd4629a1ae7630 0.1s
=> => resolve docker.io/library/ubuntu:22.04@sha256:5ec03bb3441e8b0bf3b4f9cd4629a1ae7630 0.1s
=> [internal] load build context                                       0.0s
=> => transferring context: 35B                                        0.0s
=> CACHED [2/5] RUN apt-get update && apt-get install -y --no-install-recommends 0.0s
=> CACHED [3/5] RUN useradd -m -s /bin/bash user && echo "user:user" | chpasswd 0.0s
=> CACHED [4/5] COPY entrypoint.sh /entrypoint.sh                    0.0s
=> CACHED [5/5] RUN chmod +x /entrypoint.sh                           0.0s
=> exporting to image                                                 0.3s
=> => exporting layers                                                0.0s
```

2. SSH to headnode

```
wilson@DESKTOP-KSAB409:/mnt/c/College_Computer_Science/Sem 7/high-performance-computing/assignment-3$ mkdir -p ~/.ssh
cp shared/id_ed25519 ~/.ssh/cluster_key
chmod 600 ~/.ssh/cluster_key
ssh -p 2222 -i ~/.ssh/cluster_key user@localhost
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

user@headnode:~$
```

3. Run parallel program:

```
user@headnode:~$ /shared/program.sh
/shared/program.sh
Warning: Permanently added 'computenode2' (ED25519) to the list of known hosts.
Warning: Permanently added 'computenode1' (ED25519) to the list of known hosts.
█
```

4. Parallel program output:

 runid.0.headnode runid.1.computenode1 runid.2.computenode2